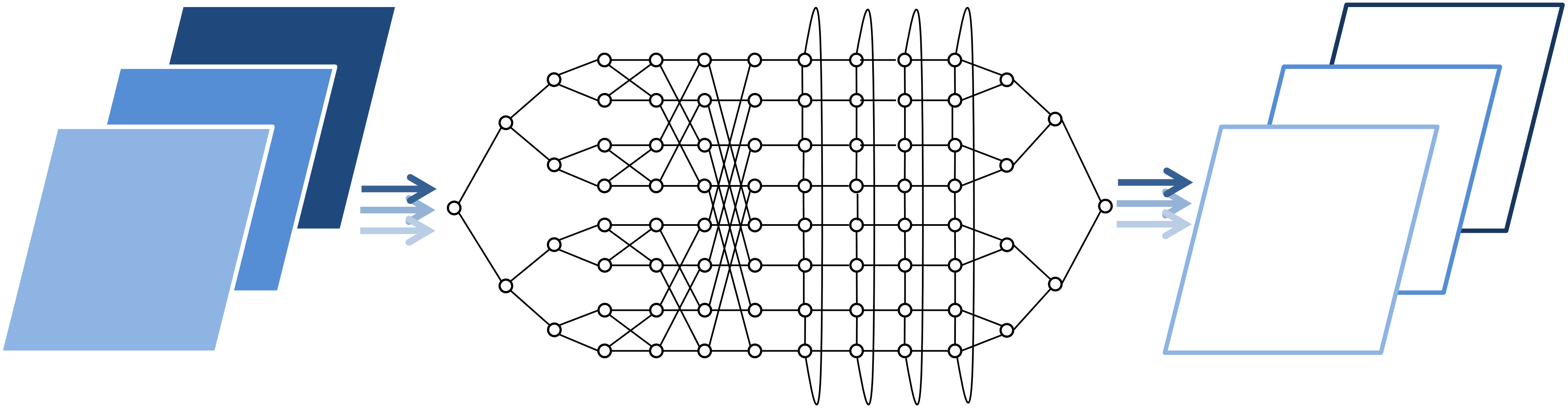
ENSIIE-HPC/BigData

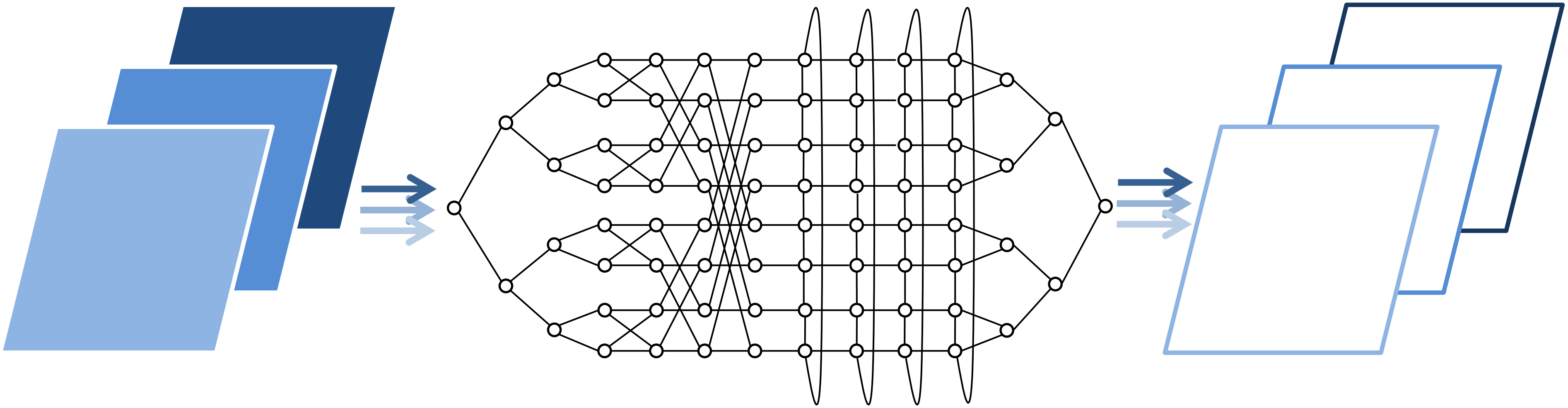
Programmation Parallèle

MPI: Message Passing Interface



Collective communications

ENSIIE-HPC/BigData-PP-IIP-Lecture 2



Collectives

- Like p2p, every collective operation requires a communicator
  - Define set of processes that will participate
- All processes in the communicator participate in the collective communication, and have to call the collective function
- All communications which are not p2p are collectives
  - p2p:
    - One-to-one
  - Collectives:
    - One-to-All
    - All-to-One
    - All-to-all

COLLECTIVE COMMUNICATION

Synchronization

synchronization

- Communication and synchronization :
  - To ensure the consistency of a calculation, parallel actions can have a « meeting point » before resuming their execution

- These « meeting points » are called **synchronization** :
  - If the synchronization concerns all parallel actions
    - Global or collective synchronization
  - If the actions have different addressing space
    - Communications

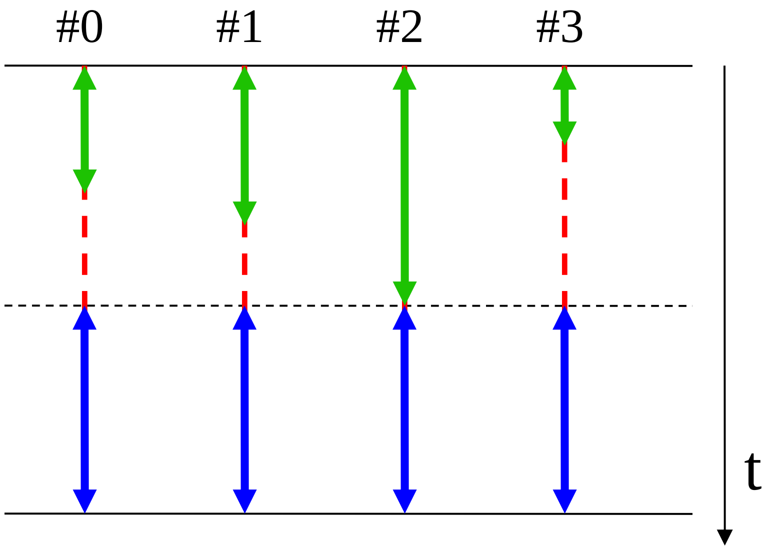
- Communications
  - Global synchronization => **global or collective communication**
  - synchronization between 2 actions => **point-to-point communication**

Barrier

- Synchronize all processes belonging to target communicator

```
int MPI_Barrier( MPI_Comm comm ) ;
```

```
MPI_Init(&argc, &argv);

/* Work 1 */

MPI_Barrier(MPI_COMM_WORLD);

/* Work 2 */

MPI_Finalize();
```



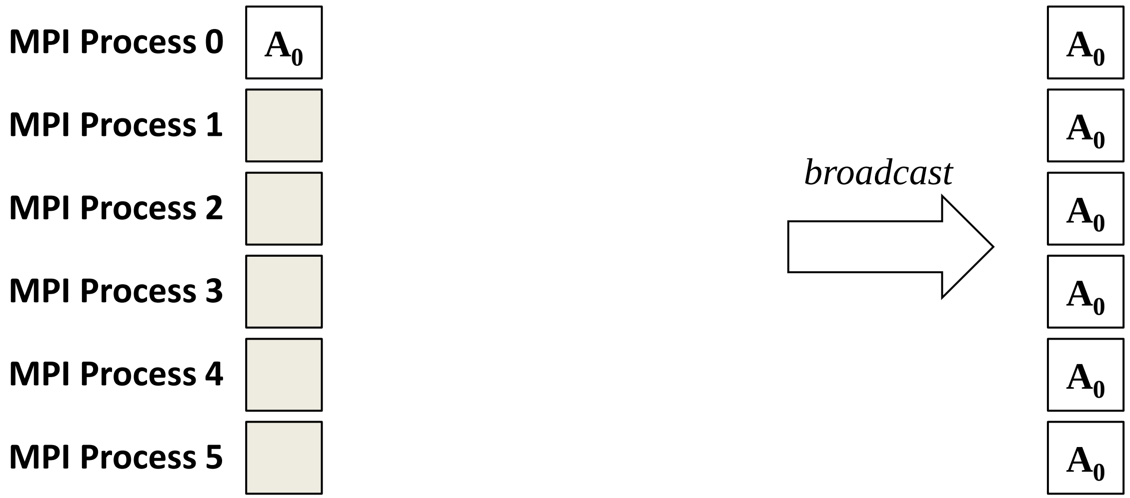
COLLECTIVE COMMUNICATION

Data Exchange

Broadcast

- Send data owned by one process to all other processes inside target communicator
- Process emitting data ➔ *root*

- One-to-all collective communication



Broadcast

```
int MPI_Bcast (
   void *buf(inout),
   int count(in),
   MPI_Datatype datatype(in),
   int root(in),
   MPI_Comm comm(in),
);
```

- rank == **root** → address of memory zone to send
- rank != **root** → address where to store broadcasted data

Output memory segment should be allocated by user.

Size of broadcasted data (number of elements of type `datatype`).

Broadcast

```
int MPI_Bcast (
  void *buf(inout),
  int count(in),
  MPI_Datatype datatype(in),
  int root(in),
  MPI_Comm comm(in),
);
```

Root rank.

This rank is valid inside `comm` communicator.

All processes involved in this collective should have the same root.

Broadcast

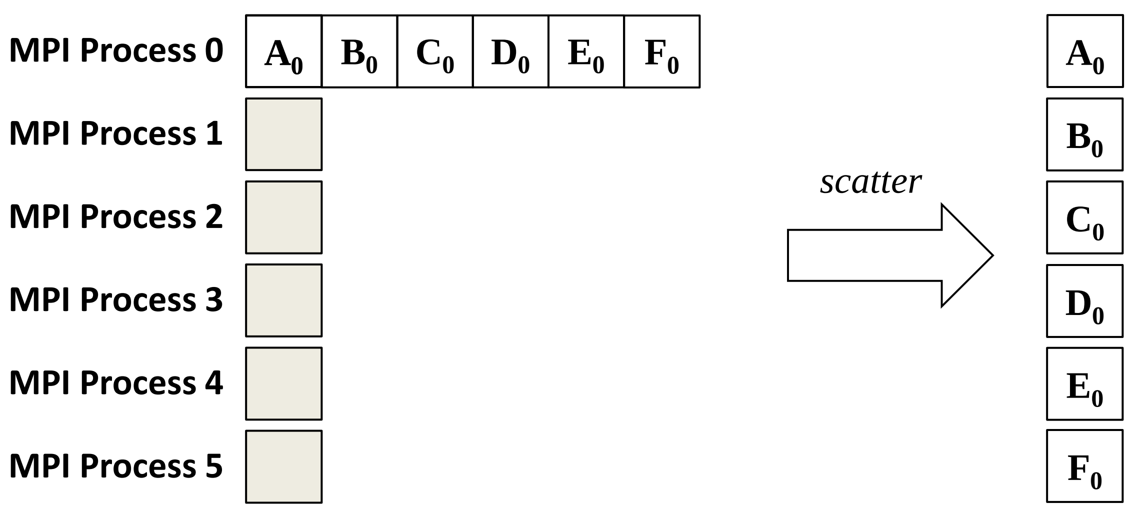
```
int me, root;
float pi = 0.0 ;

root = 0; /* Process 0 is the root */
…
MPI_Comm_rank(MPI_COMM_WORLD, &me);
…
if (me == root)
   pi = 3.14; /* Only root has the right initial value */

/* All processes have to call MPI_Bcast */
MPI_Bcast(&pi, 1, MPI_FLOAT, root, MPI_COMM_WORLD);

printf("P%d: pi = %f\n", me, pi);
```

```
% srun -n 4 ./a.out
P0: pi = 3.14
P3: pi = 3.14
P1: pi = 3.14
P2: pi = 3.14
%
```

Scatter

- One root send different data to other processes inside target communicator
- Sent data: same size and same type

- One-to-all collective communication



Scatter

```
int MPI_Scatter (
    void *sendbuf(in),
    int sendcount(in),
    MPI_Datatype sndtatyp(in),
    void *recvbuf(out),
    int recvcount(in),
    MPI_Datatype rcvtatyp(in),
    int root(in),
    MPI_Comm comm(in) );
```

Start address of memory segment to be sent to each process in `comm` communicator.

`sendbuf` is only valid for root process.

Scatter

```
int MPI_Scatter (
   void *sendbuf(in),
   int sendcount(in),
   MPI_Datatype sndtatyp(in),
   void *recvbuf(out),
   int recvcount(in),
   MPI_Datatype rcvtatyp(in),
   int root(in),
   MPI_Comm comm(in) );
```

Message size to send to each target process.

Thus, `sendbuf` array is of size `sendcount` $\times$ |`comm`| elements.

Elements to be sent to rank p start at address `sendbuf` + $p\times$ `sendcount`.

Scatter

```
int MPI_Scatter (
   void *sendbuf(in),
   int sendcount(in),
   MPI_Datatype sndtatyp(in),
   void *recvbuf(out),
   int recvcount(in),
   MPI_Datatype rcvtatyp(in),
   int root(in),
   MPI_Comm comm(in) );
```

Address where to put received data.

This memory segment should be allocated before calling the collective function.

Scatter

```
int MPI_Scatter (
   void *sendbuf(in),
   int sendcount(in),
   MPI_Datatype sndtatyp(in),
   void *recvbuf(out),
   int recvcount(in),
   MPI_Datatype rcvtatyp(in),
   int root(in),
   MPI_Comm comm(in) );
```

Size of message to receive (equals to `sendcount`).

Size is in number of elements of type `rcvtatyp`.

Scatter

```
int MPI_Scatter (
    void *sendbuf(in),
    int sendcount(in),
    MPI_Datatype sndtatyp(in),
    void *recvbuf(out),
    int recvcount(in),
    MPI_Datatype rcvtatyp(in),
    int root(in),
    MPI_Comm comm(in) );
```

Root rank.

This rank should be valid inside communicator `comm`.

All processes should have the same root.

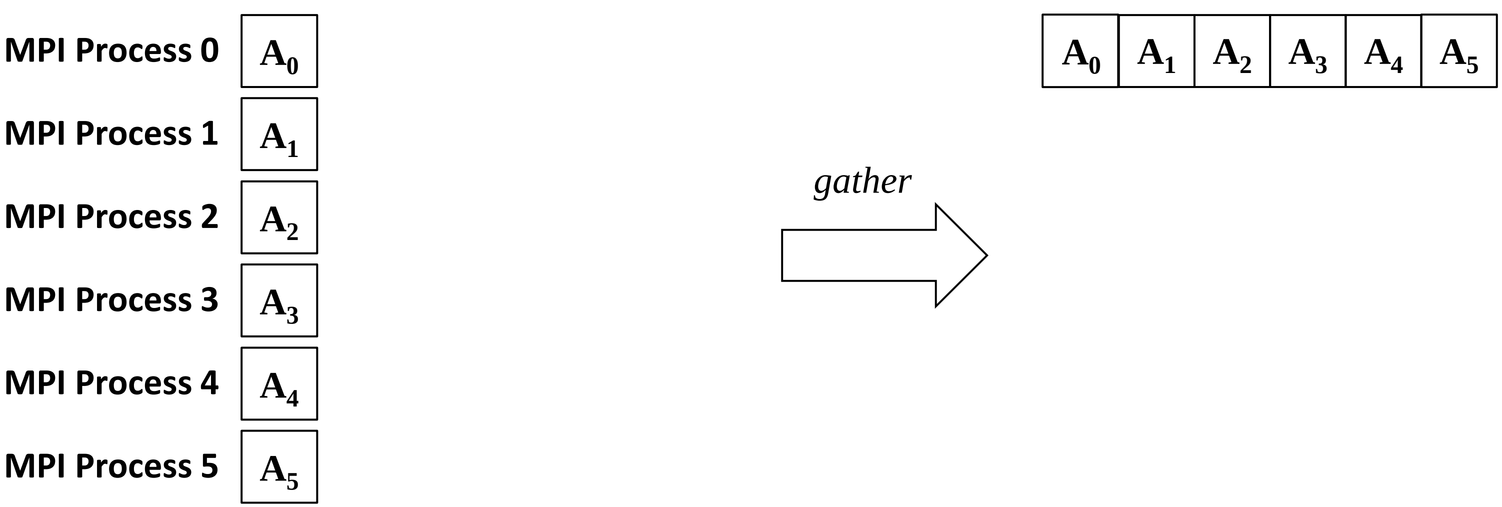
Scatter

```
int me, v, root, P;
int *sdbuf;

root = 0; /* Process 0 is the root */

MPI_Comm_rank(MPI_COMM_WORLD, &me);
MPI_Comm_size(MPI_COMM_WORLD, &P);

if (me == root) {
   sdbuf = (int*)malloc(P*sizeof(int));
   fd = fopen(« my_data", "r");
   for( p = 0 ; p < P ; p++ )
      sdbuf[p] = read_file_value(fd, p);
   fclose( fd );
} else { sdbuf = NULL; }

MPI_Scatter(sdbuf, 1, MPI_INT,
            &v, 1, MPI_INT, root, MPI_COMM_WORLD);

printf("P%d: received value = %d\n", me, v);
```

```
% cat my_data
18
25
6
3

% srun -n 4 ./a.out
P0: received value = 18
P3: received value = 3
P1: received value = 25
P2: received valud = 6
%
```

Gather

- Root process collects data from other processes (reverse of scatter)
- Sent data: same size and same type
- All-to-one collective communication



Gather

```
int MPI_Gather (
   void *sendbuf(in),
   int sendcount(in),
   MPI_Datatype sndtatyp(in),
   void *recvbuf(out),
   int recvcount(in),
   MPI_Datatype rcvtatyp(in),
   int root(in),
   MPI_Comm comm(in) );
```

Address of memory segment containing data to send to root process.

Gather

```
int MPI_Gather (
    void *sendbuf(in),
    int sendcount(in),
    MPI_Datatype sndtatyp(in),
    void *recvbuf(out),
    int recvcount(in),
    MPI_Datatype rcvtatyp(in),
    int root(in),
    MPI_Comm comm(in) );
```

Size (in number of elements of type `sndtatyp`) of message to send to root process.

Gather

```
int MPI_Gather (
    void *sendbuf(in),
    int sendcount(in),
    MPI_Datatype sndtatyp(in),
    void *recvbuf(out),
    int recvcount(in),
    MPI_Datatype rcvtatyp(in),
    int root(in),
    MPI_Comm comm(in) );
```

Address of memory segment to receive data from each process inside `comm` communicator.

Only valid for root.

Gather

```
int MPI_Gather (
    void *sendbuf(in),
    int sendcount(in),
    MPI_Datatype sndtatyp(in),
    void *recvbuf(out),
    int recvcount(in),
    MPI_Datatype rcvtatyp(in),
    int root(in),
    MPI_Comm comm(in) );
```

Size of data to receive (per process).

Array `recvbuf` must be of size `recvcount` × |`comm`| elements.

Gather

```
int MPI_Gather (
  void *sendbuf(in),
  int sendcount(in),
  MPI_Datatype sndtatyp(in),
  void *recvbuf(out),
  int recvcount(in),
  MPI_Datatype rcvtatyp(in),
  int root(in),
  MPI_Comm comm(in) );
```

Root rank.

This rank is valid inside comm communicator

All processes must use the same root.

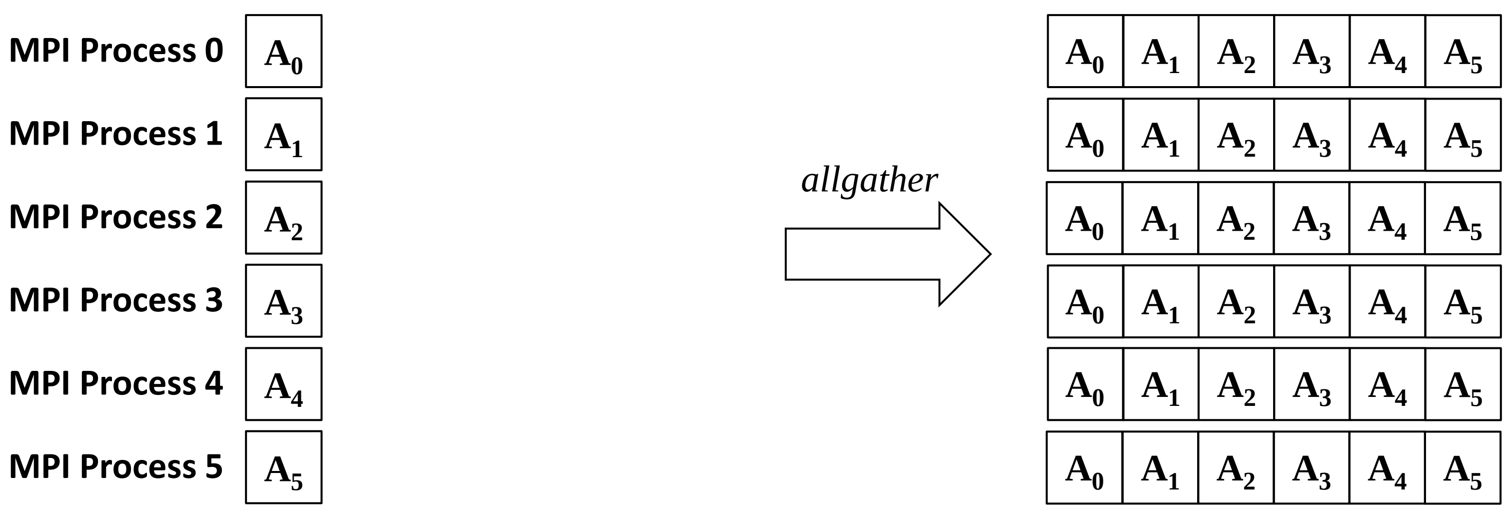
Gather

```
int me, v, root, P; int *rcvbuf;
root = 0; /* Process 0 is the root */

MPI_Comm_rank(MPI_COMM_WORLD, &me);
MPI_Comm_size(MPI_COMM_WORLD, &P);
…
if (me == root)
   rcvbuf = (int*)malloc(P*sizeof(int));
else rcvbuf = NULL;

v = … ;
MPI_Gather(&v, 1, MPI_INT,
             rcvbuf, 1, MPI_INT, root, MPI_COMM_WORLD);

if (me == root) {
   fd = fopen("results", "w");
   for( p = 0 ; p < P ; p++ )
      write_file_value(fd, p, rcvbuf[p]);
   fclose( fd );
}
```

```
% srun -n 4 a.out
% cat results
30
-100
23
19
```

Allgather

- Equivalent to gather collective operation except that every process involved inside the communication receives the final result.
- Allgather ≈ Gather + Broadcast



Allgather

```
int MPI_Allgather (
   void *sendbuf(in),
   int sendcount(in),
   MPI_Datatype sndtatyp(in),
   void *recvbuf(out),
   int recvcount(in),
   MPI_Datatype rcvtatyp(in),
   MPI_Comm comm(in),
);
```

Arguments corresponding to sent data

Arguments corresponding to received data

Alltoall

- Every Process sends and receives data
- Each process send its ith pack of data to the ith rank
- Realize a sort of matrix transpose

MPI Process 0	A_0	B_0	C_0	D_0	E_0	F_0
MPI Process 1	A_1	B_1	C_1	D_1	E_1	F_1
MPI Process 2	A_2	B_2	C_2	D_2	E_2	F_2
MPI Process 3	A_3	B_3	C_3	D_3	E_3	F_3
MPI Process 4	A_4	B_4	C_4	D_4	E_4	F_4
MPI Process 5	A_5	B_5	C_5	D_5	E_5	F_5

alltoall

A_0	A_1	A_2	A_3	A_4	A_5
B_0	B_1	B_2	B_3	B_4	B_5
C_0	C_1	C_2	C_3	C_4	C_5
D_0	D_1	D_2	D_3	D_4	D_5
E_0	E_1	E_2	E_3	E_4	E_5
F_0	F	F_2	F_3	F_4	F_5

Alltoall

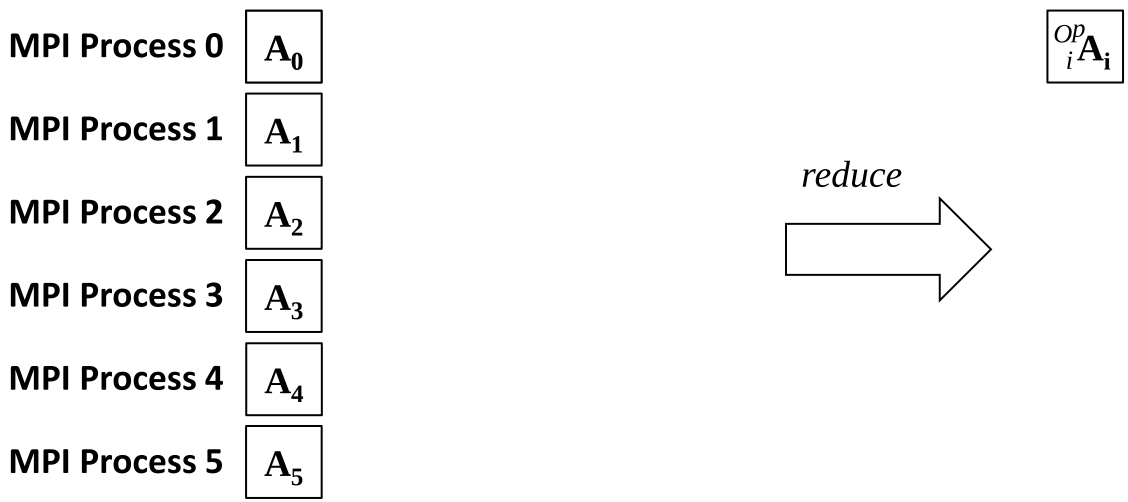
```
int MPI_Alltoall (
   void *sendbuf(in),
   int sendcount(in),
   MPI_Datatype sendtype(in),
   void *recvbuf(out),
   int recvcount(in),
   MPI_Datatype recvtype(in),
   MPI_Comm comm(in),
);
```

Arguments corresponding to sent data

Arguments corresponding to received data

Reduction

- Root rank collects data from other processes and apply one specific operation
- MPI allows multiple reductions at the same time



Reduction

```
int MPI_Reduce (
   void *sendbuf(in),
   void *recvbuf(out),
   int count(in),
   MPI_Datatype datatype(in),
   MPI_Op op(in),
   int root(in),
   MPI_Comm comm(in),
);
```

Address of data to send to root process.

Array of `count` elements of `datatype` type.

Reduction

```
int MPI_Reduce (
   void *sendbuf(in),
   void *recvbuf(out),
   int count(in),
   MPI_Datatype datatype(in),
   MPI_Op op(in),
   int root(in),
   MPI_Comm comm(in),
);
```

Address of final data (after reduction).

Only valid for root process.

Array of `count` elements of `datatype` type.

Reduction

```
int MPI_Reduce (
   void *sendbuf(in),
   void *recvbuf(out),
   int count(in),
   MPI_Datatype datatype(in),
   MPI_Op op(in),
   int root(in),
   MPI_Comm comm(in),
);
```

Operation to perform on each element of every process.

Upon return, root will have:
`recvbuf`[i] = $\texttt{op}_{0 \leq p < P}$ `sendbuf`$_p$ [i]
with $0 \leq i <$ `count`
and P size of `comm`.

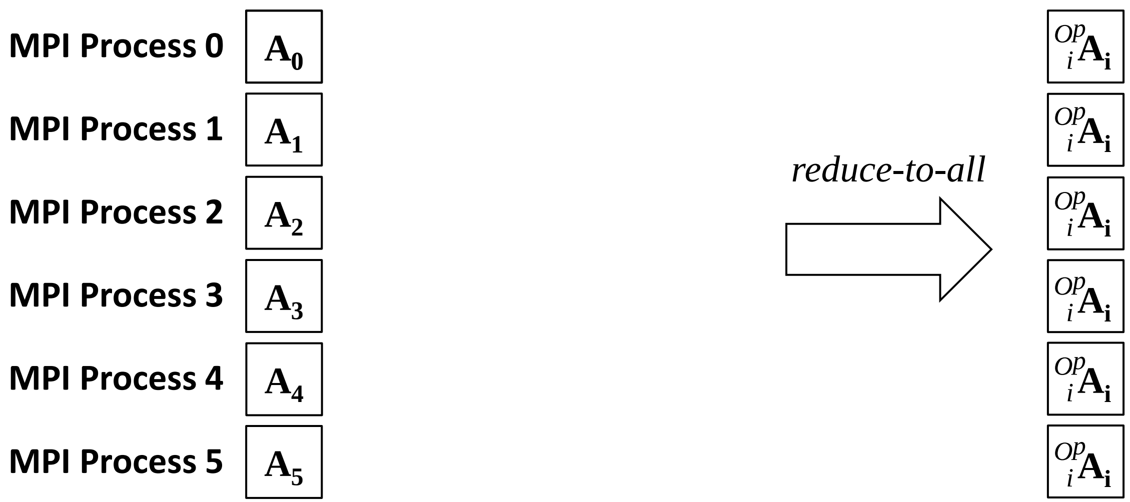
Reduction

- Reduction operations are of type `MPI_Op`
- Possibility to create user-defined ones: `MPI_Op_create()`

MPI Operation	***Meaning***	***Corresponding Type***
MPI_MAX	Maximum	Integers and real
MPI_MIN	Minimum	Integers and real
MPI_SUM	Sum	Integer and real
MPI_PROD	Product	Integer and real
MPI_LAND	Logical AND	Integer
MPI_BAND	Binary AND	Integer and MPI_BYTE
MPI_LOR	Logical OR	Integer
MPI_BOR	Binary OR	integer MPI_BYTE
MPI_LXOR	Logical XOR	Integer
MPI_BXOR	Binary XOR	Integer and MPI_BYTE
MPI_MAXLOC	Maximum w/ index	Structures w/ 2 integers
MPI_MINLOC	Minimum w/ index	Structures w/ 2 integers

Reduce to All

- Equivalent to a reduction
- But every process has the final result
- No need to specify a root process
- Allreduce ≈ Reduce + Broadcast



Reduce to All

```
int MPI_Allreduce (
  void *sendbuf(in),
  void *recvbuf(out),
  int count(in),
  MPI_Datatype datatype(in),
  MPI_Op op(in),
  MPI_Comm comm(in),
);
```

Same signature as `MPI_Reduce` but no root is needed.

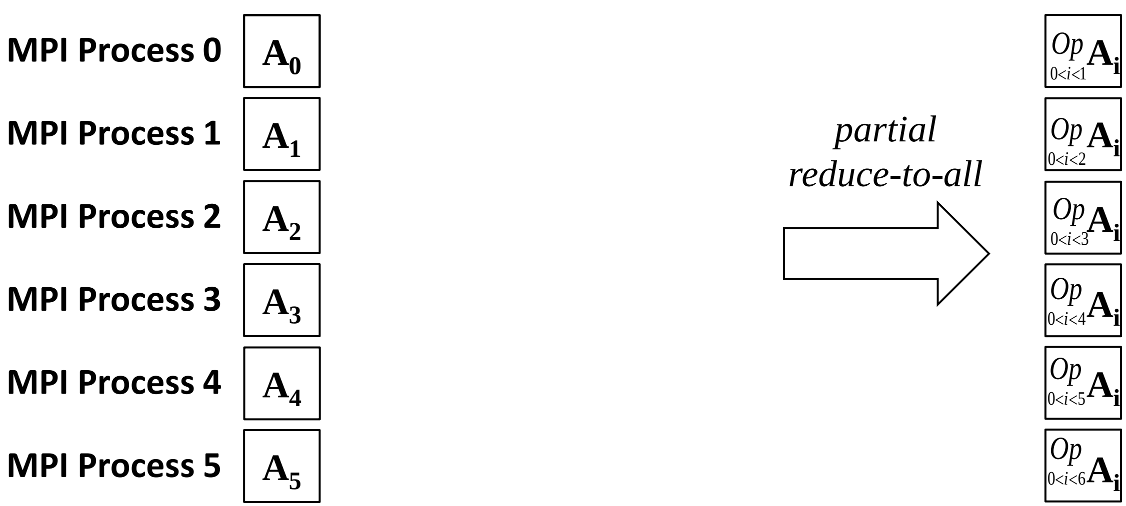
Reduce to All

```
% srun -p 4 ./a.out
1+…+100 = 5050
1+…+100 = 5050
1+…+100 = 5050
1+…+100 = 5050
%
```

```
int P, N = 100;
int me, i, sum, sum_loc = 0;

MPI_Comm_rank(MPI_COMM_WORLD, &me);
MPI_Comm_size(MPI_COMM_WORLD, &P);
…
/* Works if N%P == 0 */
for( i = 1 + me*N/P ; i <= (me+1)*N/P ; i++ )
   sum_loc += i;

MPI_Allreduce(&sum_loc, &sum_glob, 1, MPI_INT,
              MPI_SUM, MPI_COMM_WORLD);

/* After reduction, all processes have, in sum_glob, the sum of N first integers
*/

printf("1+…+%d = %d\n", N, sum_glob);
```

Partial Reduction

- All ranks collect data from other processes **with smaller rank number** and apply one specific operation, including its contribution
- MPI allows multiple reductions as the same time



Partial Reduction

```
int MPI_Scan (
   void *sendbuf(in),
   void *recvbuf(out),
   int count(in),
   MPI_Datatype datatype(in),
   MPI_Op op(in),
   MPI_Comm comm(in),
);
```

Same signature as MPI_Allreduce

COLLECTIVE COMMUNICATIONS

Multiple Sizes

Per-Rank Data Size

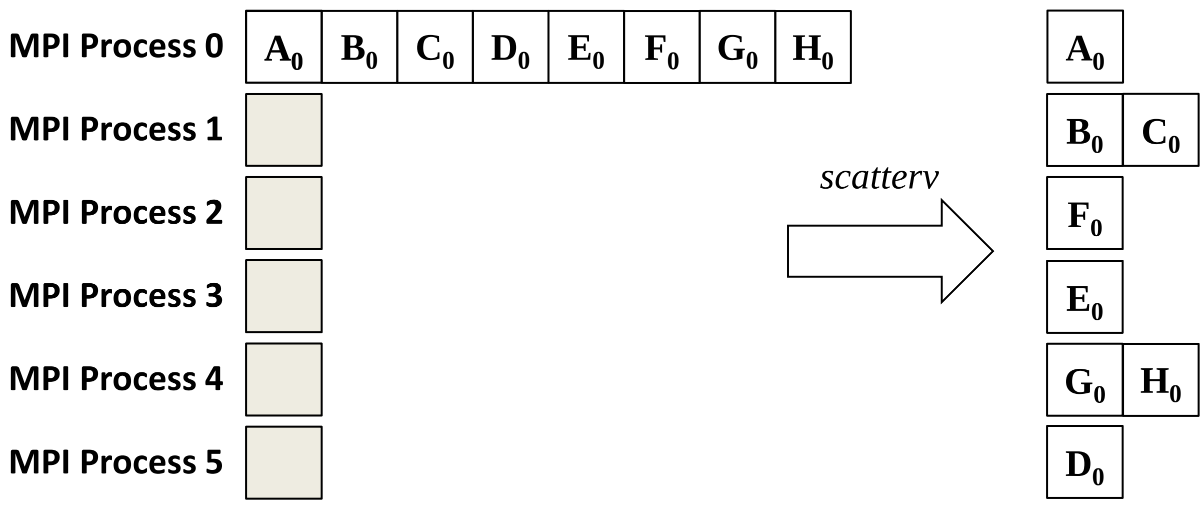
- Some collective communications propose multiple versions including one to handle different size for different ranks.
  - E.g., Broadcast, Gather

- Corresponding names have the suffix `v`
  - `v` = varying or vector or variant

- Examples
  - `MPI_Gather` → `MPI_Gatherv`
  - `MPI_Allgather` → `MPI_Allgatherv`
  - `MPI_Scatter` → `MPI_Scatterv`
  - `MPI_Alltoall` → `MPI_Alltoallv`

MPI_Scatterv

- Data A_i have the same type
- Unlike `MPI_Scatter`, A_0, A_1... may not have the same size
  - Need another argument to specify size of each data
- Unlike `MPI_Scatter`, data Ai may not be contiguous (in memory)
  - Need another argument to specify base address of each data



MPI_Scatterv

```
int MPI_Scatterv (
   void *sendbuf(in),
   int *sendcounts(in),
   int *displs(in),
   MPI_Datatype sendtype(in),
   void *recvbuf(out),
   int recvcount(in),
   MPI_Datatype recvtype(in),
   int root(in), MPI_Comm comm(in) );
```

sendbuf = array of data to distribute to each process inside communicator comm.

sendcounts[p] elements have to send to process p ($0 \leq p < P$).
Those elements are store at address

sendbuf + displs[p] .

All elements are of type sendtype.

Those arguments are only valid for root.

MPI_Scatterv

```
int MPI_Scatterv (
    void *sendbuf(in),
    int *sendcounts(in),
    int *displs(in),
    MPI_Datatype sendtype(in),
    void *recvbuf(out),
    int recvcount(in),
    MPI_Datatype recvtype(in),
    int root(in), MPI_Comm comm(in) );
```

recvbuf = array of data to receive.

Array with recvcount elements of type rcvtatyp.

MPI_Scatterv

File `data`

Per-process data size
N_1
N_2
…
N_P

Data for process of rank 0
val^0_0
val^0_1
…
$val^0_{N_0}$

Data for process of rank 1
val^1_0
val^1_1
…
$val^1_{N_1}$

…

Data for process of rank P
val^P_0
val^P_1
…
$val^P_{N_P}$

- One process reads a file named data
- This file contains data for each process
- File format is described on the left-hand-side
  - Data size for each process
  - Data (e.g., float values) stored in a per-process manner
- After parsing step, all processes get their own piece of data

MPI_Scatterv

```
int sz; int *szbuf, *displs;
double *sdbuf, *rcvbuf;

if (me == root) {
   szbuf = (int*)malloc(P*sizeof(int));
   displs = (int*)malloc((P+1)*sizeof(int));
   fd = fopen("data", "r");
   displs[0] = 0;
   for( p = 0 ; p < P ; p++ ) {
      szbuf[p] = read_int_value(fd, p);
      displs[p+1] = displs[p] + szbuf[p];
   }
   sdbuf = (double*)malloc(displs[P]*sizeof(double));
   for( p = 0 ; p < P ; p++ )
       for( i = 0 ; i < szbuf[p] ; i++ )
           sdbuf[displs[p]+i] = read_float_value(fd, p);
   fclose( fd );
}
```

```
/* Read data sizes for each
process (szbuf array)
Compute corresponding offset
(displs array) */
```

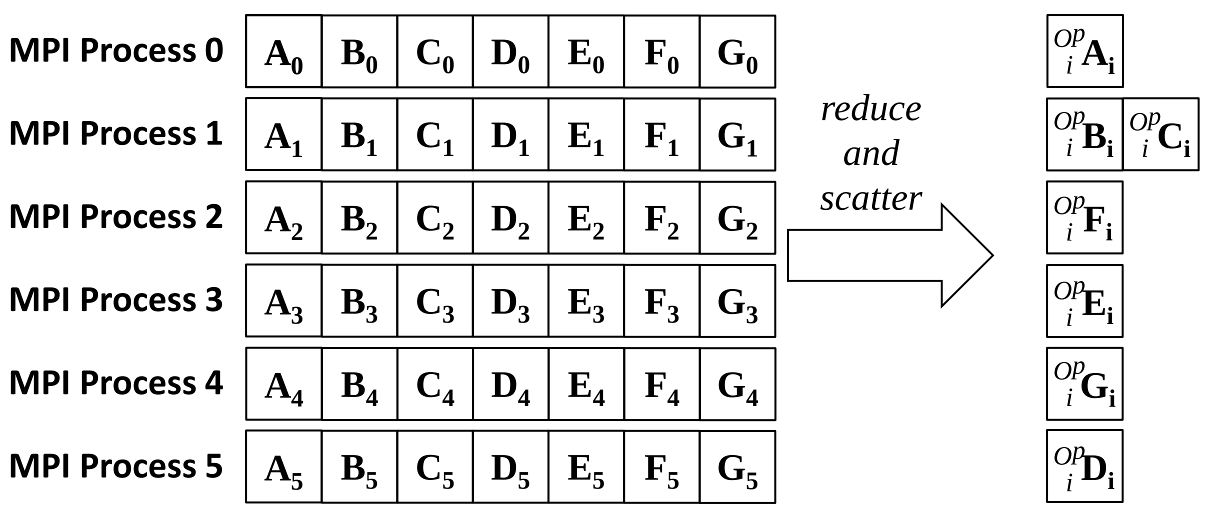
```
/* Read per-block
data (sdbuf array).
Blocks will be
distributed amon
ranks */
```

MPI_Scatterv

```
else {
   szbuf = displs = NULL;
   sdbuf = NULL;
}
/* All processes must call MPI_Scatter
   Corresponding data size is stored into sz
 */
MPI_Scatter(szbuf, 1, MPI_INT,
            &sz, 1, MPI_INT,
            root, MPI_COMM_WORLD);

rcvbuf = (double*)malloc(sz*sizeof(double));
/* All process must call MPI_Scatterv
   Corresponding data are stored in rcvbuf
 */
MPI_Scatterv(sdbuf, szbuf, displs, MPI_DOUBLE,
             rcvbuf, sz, MPI_DOUBLE,
             root, MPI_COMM_WORLD);
```

Reduce-Scatter

- Combine a reduction and a scatterv
- Reduction on an array
- Every process gets a piece of array
- Like `MPI_Scatterv`, A_0, A_1… may not have the same size on the scatter part

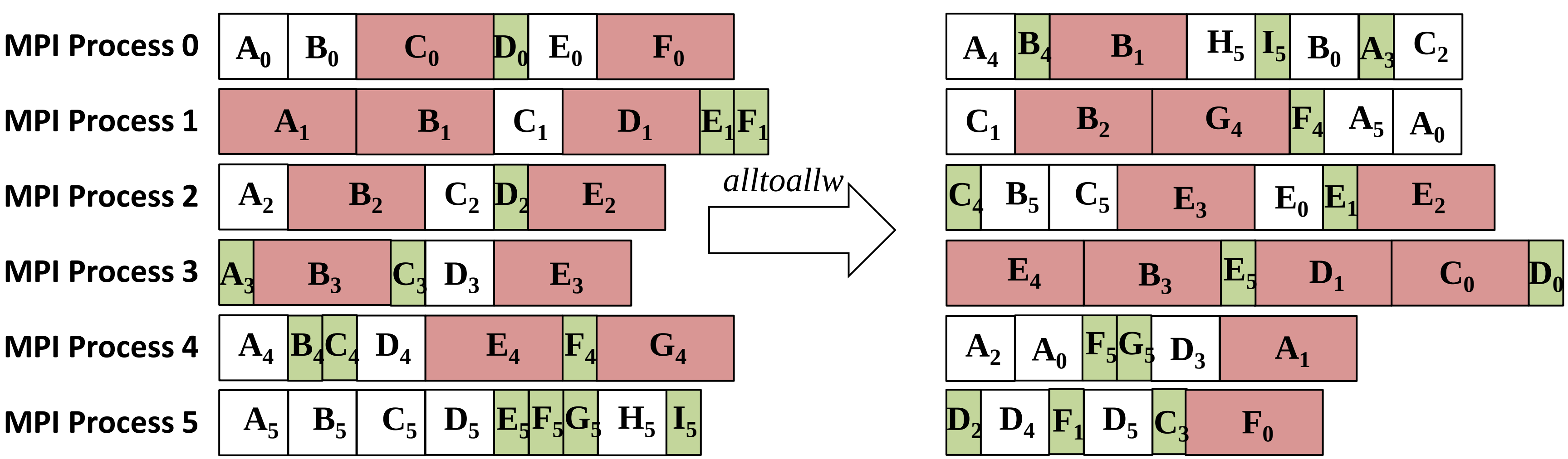


THE THREE MISTAKES

Alltoallw, Exscan, Scatter_Reduce_block

Alltoallw

- Like `MPI_Scatterv`, A_0, A_1... may not have the same size
  - Need another argument to specify size of each data
- Like `MPI_Scatterv`, data Ai may not be contiguous (in memory)
  - Need another argument to specify base address of each data
- In addition, data A_i may not have the same type
  - Datatype argument is now an array



MPI_Alltoallw

```
int MPI_Alltoallw (
    void *sendbuf(in),
    int *sendcounts(in),
    int *displs(in),
    MPI_Datatype *sendtypes(in),
    void *recvbuf(out),
    int *recvcounts(in),
    MPI_Datatype *recvtypes(in),
    int root(in), MPI_Comm comm(in) );
```

sendbuf = array of data to distribute to each process inside communicator comm.

sendcounts[p] elements have to send to process p ($0 \leq p < P$).
Those elements are store at address

sendbuf + displs[p] .

sendtype[p] is the type of elements to send to process p

MPI_Alltoallw

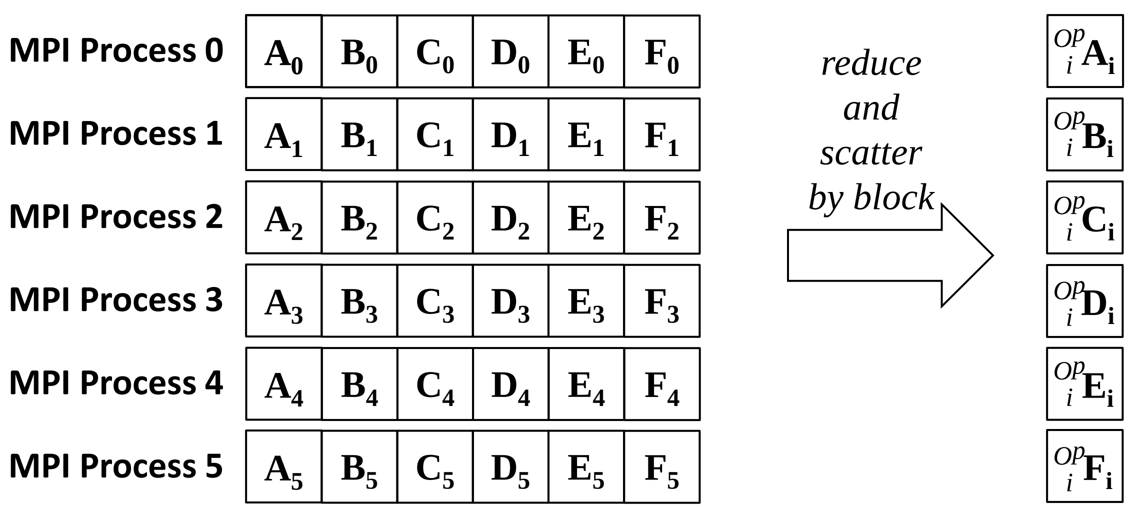
```
int MPI_Alltoallw (
    void *sendbuf(in),
    int *sendcounts(in),
    int *displs(in),
    MPI_Datatype *sendtypes(in),
    void *recvbuf(out),
    int *recvcounts(in),
    MPI_Datatype *recvtypes(in),
    int root(in), MPI_Comm comm(in) );
```

recvbuf = array of data to receive.

Array composed of recvcount[*p*] elements of type recvtype[*p*] coming from all other processes *p*.

Reduce-Scatter block

- Combine a reduction and a scatter
- Reduction on an array
- Every process gets a piece of array of same size
- Data chunk are contiguous (no displacement arg)



Reduction-Scatter block

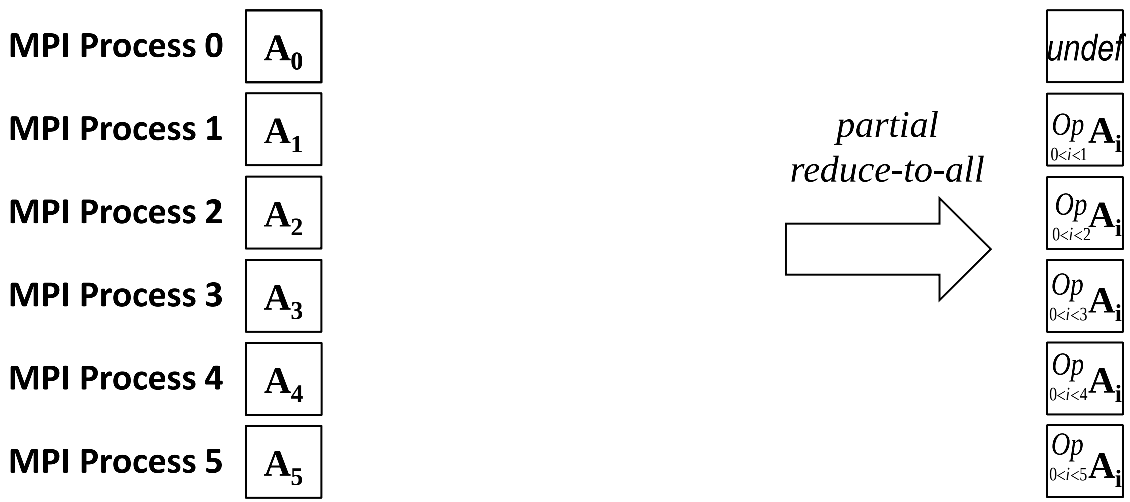
```
int MPI_Reduce_Scatter_block (
   void *sendbuf(in),
   void *recvbuf(out),
   int recvcount(in),
   MPI_Datatype datatype(in),
   MPI_Op op(in),
   MPI_Comm comm(in),
);
```

Address of data to send to root process.

Fixed `recvcount` elements of `datatype` type.

Partial Reduction

- All ranks collect data from other processes **with smaller rank number** and apply one specific operation, **excluding** its contribution
- MPI allows multiple reductions as the same time



Partial Reduction

```
int MPI_Exscan (
   void *sendbuf(in),
   void *recvbuf(out),
   int count(in),
   MPI_Datatype datatype(in),
   MPI_Op op(in),
   MPI_Comm comm(in),
);
```

Same signature as MPI_Allreduce

Why mistakes

- Reduce_scatter_block is the simplified version of reduce_scatter
  - should have been the first one to appear
  - For coherency in naming convention, reduce_scatter_block should be reduce_scatter, and reduce_scatter should be reduce_scatterv
- MPI_Scan is just (MPI_Exscan + local variable)
  - Do not need to create two functions
  - Function MPI_Exscan is enough. Should have been the first one
- MPI_Alltoallw
  - Allows the user to exchange anything
  - Multiple datatypes -> complex internal implementation and algorithms
  - Why is it the only collective to allow multiple datatypes?

COLLECTIVES

Usage

Collectives

- Completing a collective call on a process does not mean that the collective is done for all processes
  - It just means that the contribution of the current process is done
- If the current process does not require the result but only send data, it basically is the same as MPI_Send
  - After the collective call, the user can reuse the input buffer
  - It does not even mean that the data has been send
- If all processes provide input data and require an output result, the collective is synchronizing
  - However, Barrier is the only official synchronizing collective in the API

Collectives

- Every collective operation requires a communicator
  - Define set of processes that will participate
- What happen if two ranks call a different collective operation?
- What happen if two ranks in different communicator call the same collective communication at the same time?

Rules

- All processes in the group identified by the communicator must call the collective routine
- Inside a communicator, processes should call the same sequence of collective communication
- Between two ranks in different communicators, there are no restrictions

Examples

```
void f ( int r ) {
         if( r == 0 )
            MPI_Barrier(MPI_COMM_WORLD);
         return;
}
```

May be wrong (depends on value of r)

```
void g ( int r ) {
         if( r == 0 )
            MPI_Barrier(MPI_COMM_WORLD);
         else
            MPI_Barrier(MPI_COMM_WORLD);
         return;
}
```

Correct

```
void h ( int r ) {
         if( r == 0 ) {
            MPI_Reduce(MPI_COMM_WORLD, …);
            MPI_Barrier(MPI_COMM_WORLD);
         } else {
            MPI_Barrier(MPI_COMM_WORLD);
            MPI_Reduce(MPI_COMM_WORLD, …);
         }
         return; }
```

May be wrong (depends on value of r)

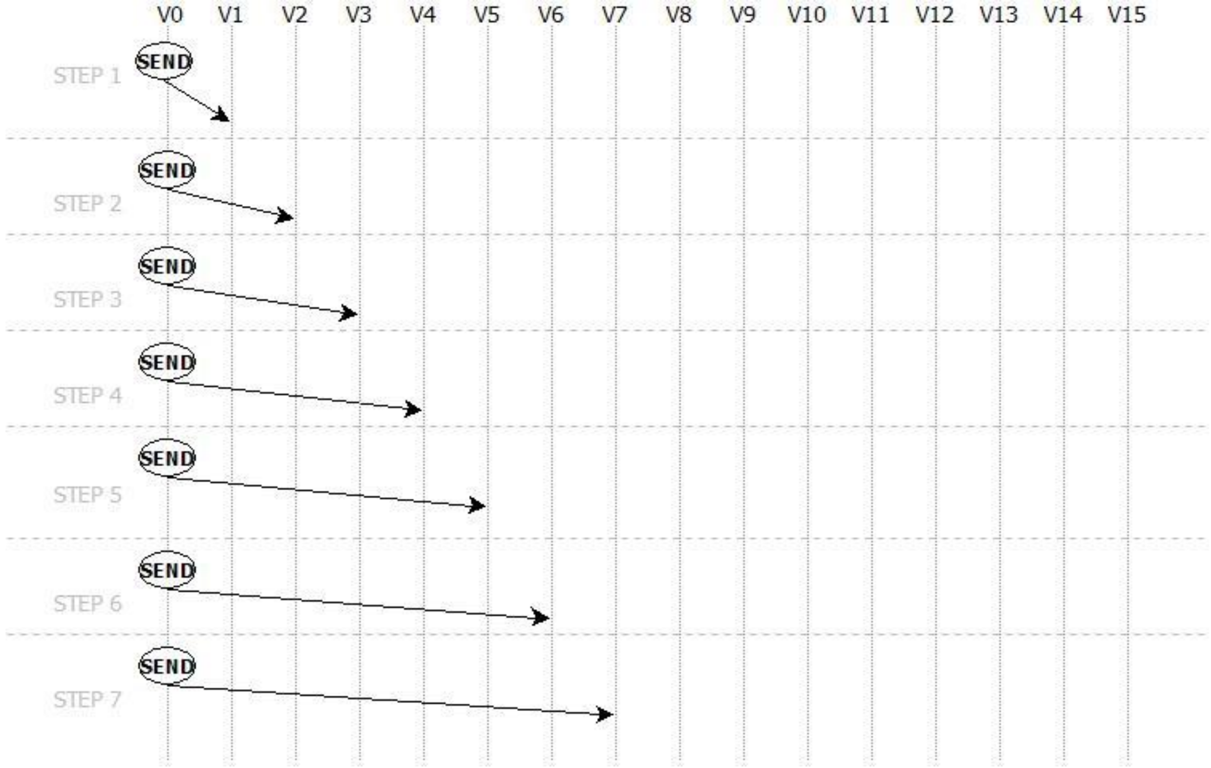
Hints

- Be careful of arguments to MPI collective communications
  - Same root
  - Compatible count for elements to send and to receive
- Be careful of control flow inside the program
  - Not related to lexical order
  - Related to dynamic order
- Be careful of communicators

COLLECTIVE ALGORITHMS

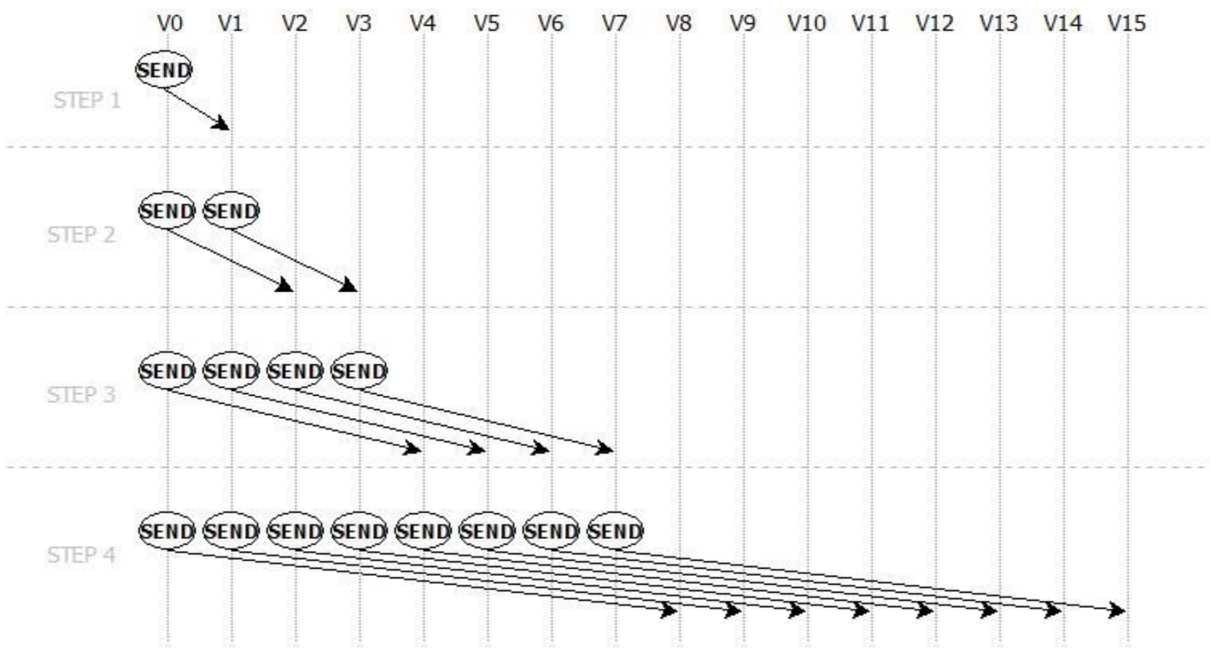
Broadcast algorithms

- Linear algorithm
- Root send its data to each other process
- Iterative and linear: for p processes, need p-1 step



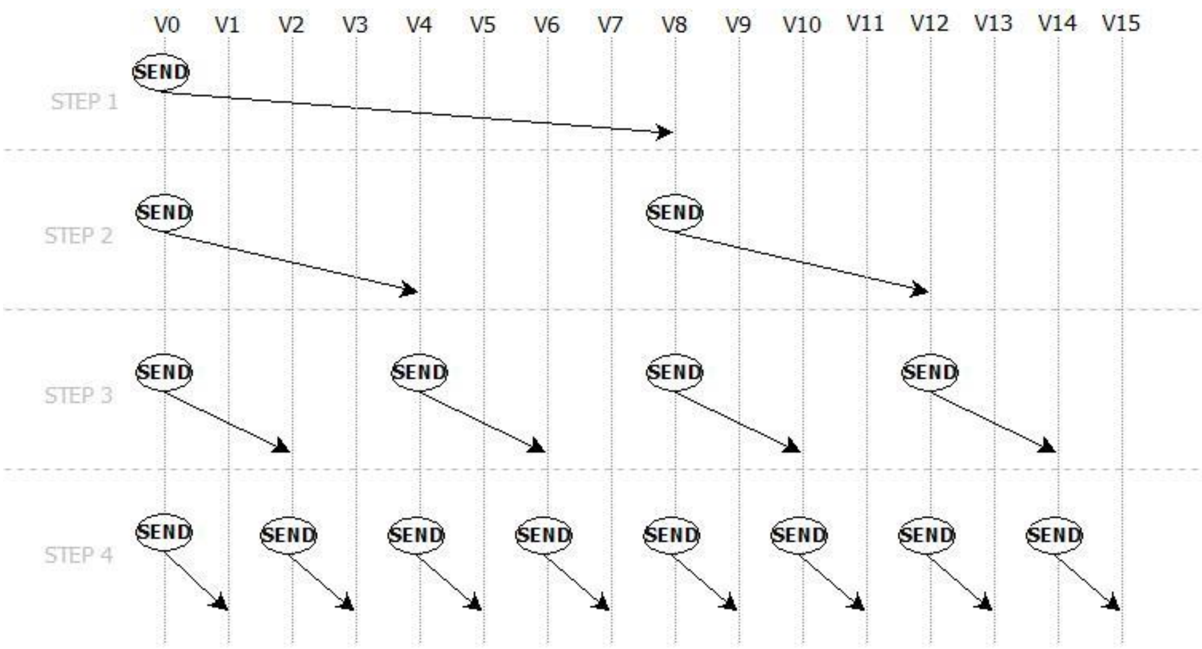
Broadcast algorithms

- Binary tree: at each step, each process with the data to broadcast send the data to another process
- Need log(p) steps
- For(0; i<log(p); i++) if(rank<=i) send(data, rank+2^i)



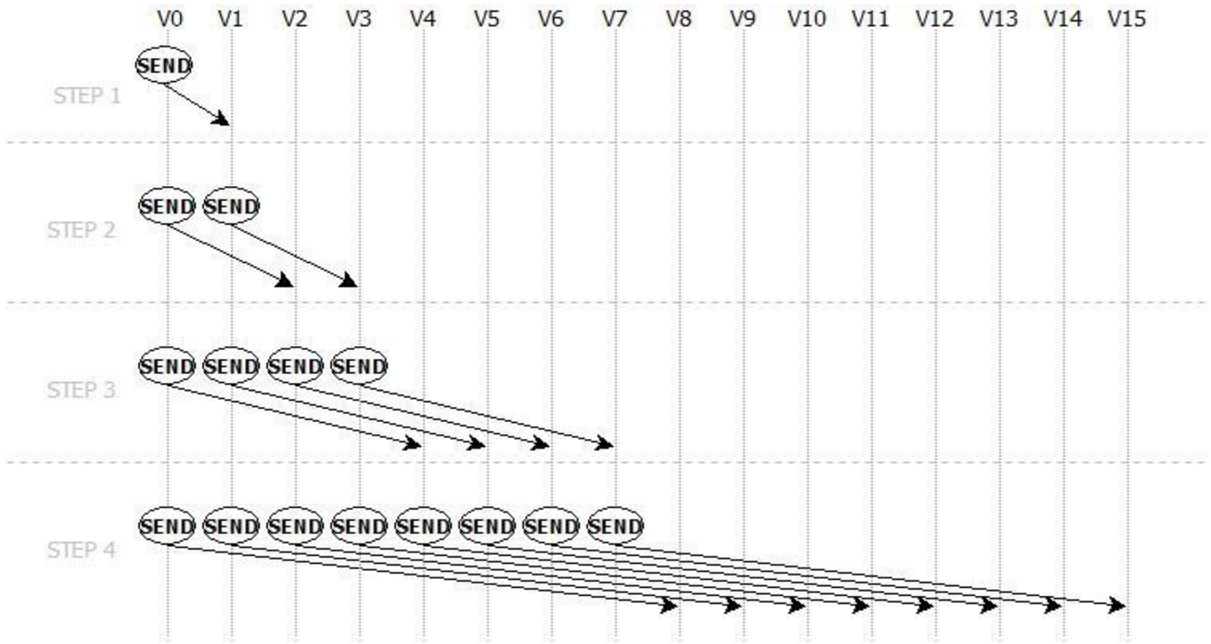
Broadcast algorithms

- Binary tree
- For(log(p); i>0; i--) if(rank%2^i==0) send(data, rank+2^{i-1})
- Are those two algorithms equivalent?



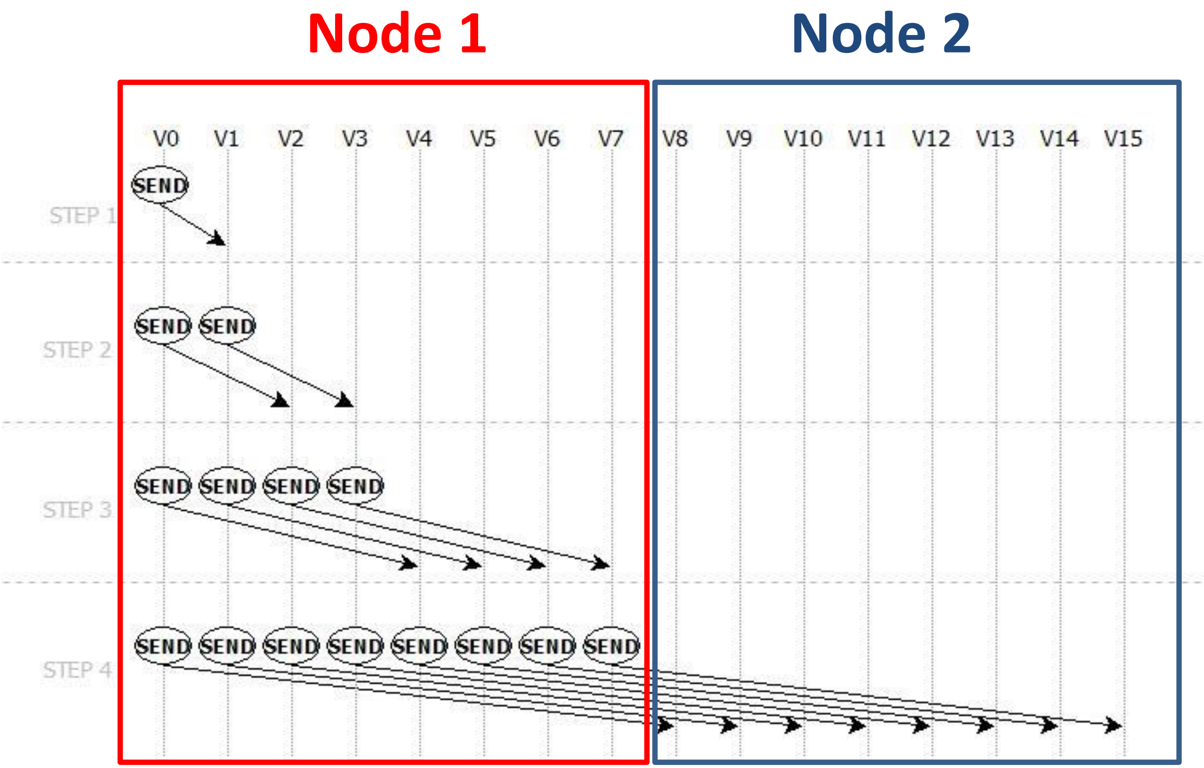
Broadcast algorithms

- If we consider two nodes



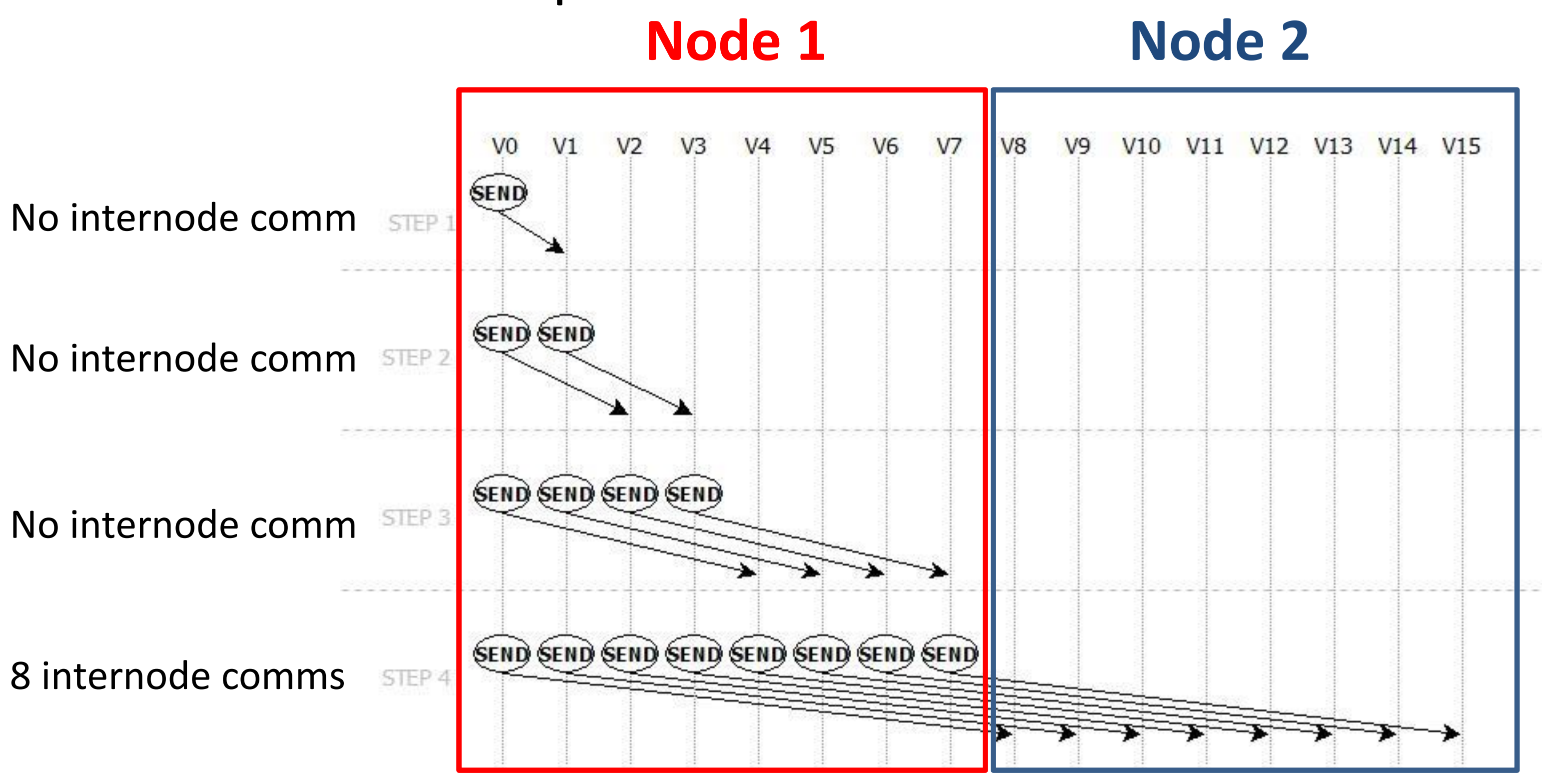
Broadcast algorithms

- If we consider two nodes



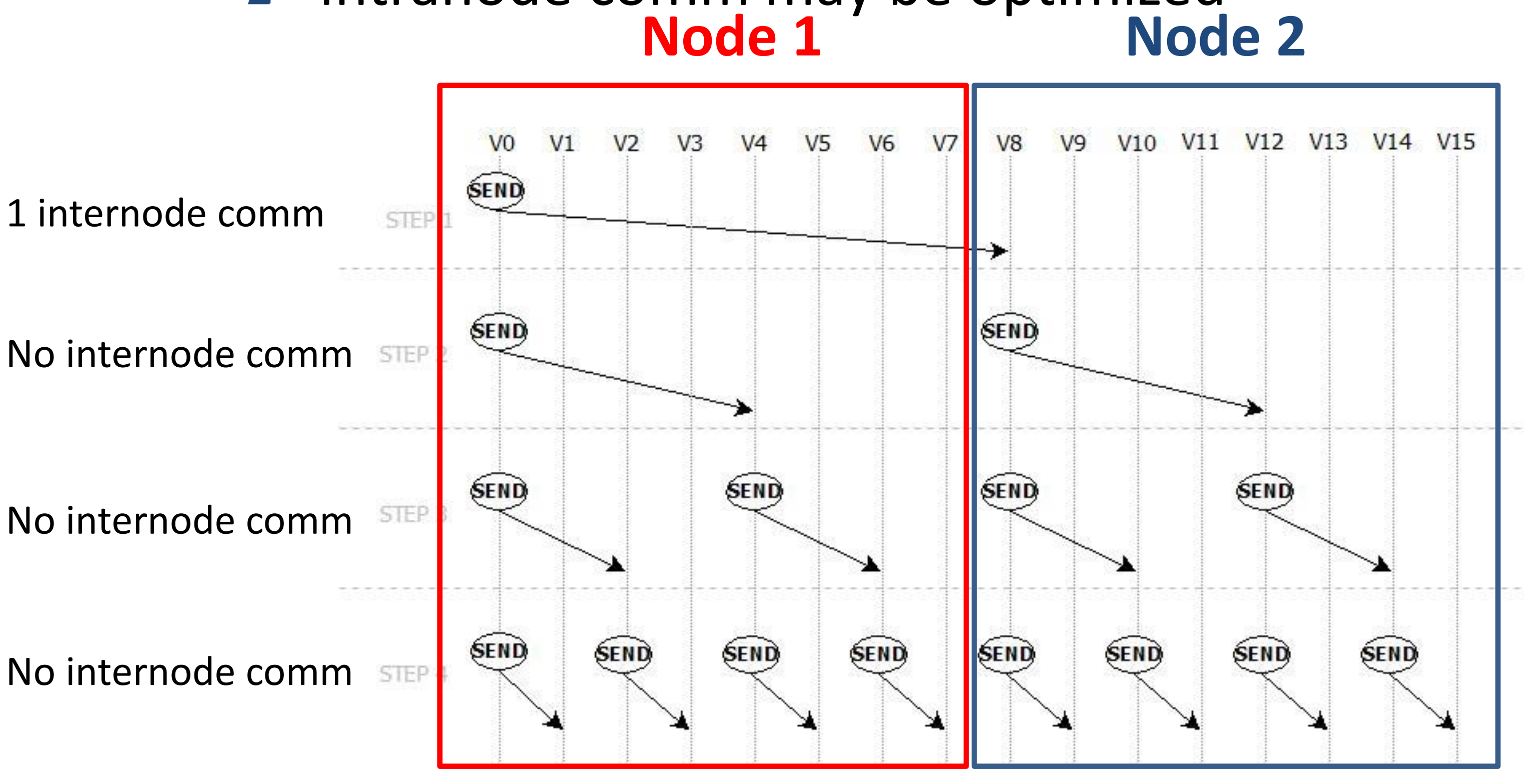
Broadcast algorithms

- If we consider two nodes
- All internode communications (network) happens at the same step



Broadcast algorithms

- Only one internode communication
- No contention on the newtwork
- Intranode comm may be optimized



Collective algorithms

- Several algorithms may be implemented for one collective in a runtime
- Default selection of algorithms often depends on the size of the messages, and the number of MPI processes involved
- Algorithm selection may also be available to user through launch options, environment variables or communicator attributes